

IILM UNIVERSITY, GREATER NOIDA

INTERSHIP REPORT PRESENTATION

DecodeLabs AI-ML

Virtual Internship

DecodeLabs Virtual Internship Program | Artificial Intelligence (AI)

Vansh

Roll No. 25SCS1003000858 | B.Tech CSE | Batch 2025–2029

July – August 2026



Vansh

B.Tech – Computer Science
& Engineering

ABOUT THE STUDENT

Candidate Profile



Roll Number

25SCS1003000858



Institute

IILM University, Greater Noida, U.P.



Programme

B.Tech CSE, Batch 2025–2029



Internship Domain

Artificial Intelligence (AI) / AI-ML



Duration

4 Weeks | 3 July – 3 August 2026

ABOUT THE PROGRAM

DecodeLabs Virtual Internship Program

DecodeLabs is a government-registered digital learning and technology enterprise operating under the theme "Your Digital Lab". It runs structured Virtual Internship Programs designed to provide engineering students with industry-aligned, hands-on exposure in emerging domains such as Artificial Intelligence and Machine Learning. The programs emphasize real-world projects, collaborative tasks, mentor guidance, and professional growth.



Govt. Registered

Government-registered enterprise focused on digital learning & technology skills.



Hands-on Projects

Real-world projects, collaborative tasks, and mentor-led sessions for practical growth.



AI / ML Track

Industry-aligned curriculum covering computer vision, CNNs, and model integration.

PROBLEM STATEMENT

Bridging Theory and Applied AI / Computer Vision

Although many engineering students learn the theoretical foundations of machine learning and neural networks in coursework, very few get structured, hands-on exposure to building and applying AI models using industry-relevant tools and workflows. Concepts such as convolutional neural networks, object detection, and image classification are often understood only theoretically — with little practice converting and integrating such models into real applications.

This internship addressed that gap — practically learning the full AI-ML workflow:



PROJECT OBJECTIVES

What the Internship Set Out to Achieve



Build ML Models

Understand ML from first principles and build models for image recognition & classification.



Object Detection

Learn basics of object detection and integrate pretrained detectors into applications.



Product Image Search

Build visual product search features across static images and live camera feeds.



Backend Integration

Understand backend integration concepts for AI-powered mobile or web applications.



Image Classification

Build, evaluate and improve custom image-classification models.



App Integration

Create and integrate a custom image classifier into a working application.

The Toolkit Behind the Internship



Python

Core programming language



TensorFlow / Keras

Model building & training



TF Lite / Lightweight

On-device / edge deployment



CNNs

Image recognition models



Object Detection

Pretrained & custom
detectors



Product Image Search

Visual search integration



Image Classification

Custom classifiers



DecodeLabs Platform

Mentor sessions &
assessments

From Trained Model to Application



Model Layer

Neural network (CNN) built & trained for image recognition



Conversion / Optimization

Models converted to lightweight formats for efficient inference



Detection Layer

Pretrained & custom object detectors on static / live feeds



Search Layer

Product image search with backend integration



Classification Layer

Custom image classifier refines results



Application Layer

Models integrated into mobile / web application

The Learning Journey (4 Weeks)

1

Foundations of ML & Neural Networks

First principles of ML through building models for image recognition & classification.

2

Object Detection Basics

Basics of object detection and integrating pretrained detectors into applications.

3

Custom Models & Product Image Search

Training custom models and building product image search on static & live feeds.

4

Integration & Classification

Backend integration, custom classifiers, and integrating a classifier into an application.

EXPECTED OUTCOMES

Skills & Capabilities Gained

- ✓ Ability to explain and apply first-principles ML to build models for image-related tasks
- ✓ Practical understanding of object detection with pretrained & custom-trained models
- ✓ Hands-on exposure to building a product image search feature with application integration
- ✓ Capability to build, evaluate and improve custom image-classification models
- ✓ Skill to integrate trained models into an application context
- ✓ Successful completion of DecodeLabs Virtual Internship in AI with verified Certificate

CERTIFICATION

Successfully Completed

Program

DecodeLabs Virtual Internship – AI

Student ID

AI086328

Duration

3 July – 3 August 2026

Date of Issue

4 August 2026

Mode

Remote / Virtual

Certificate of Completion

Awarded by DecodeLabs for successful completion of the Virtual Internship Program in Artificial Intelligence (AI), demonstrating dedication, consistency, and hands-on problem-solving skills across real-world projects and collaborative tasks.

Position

AI Intern

Track

**Artificial
Intelligence**

Offer Date

3 July 2026

Completion

Certificate + LOR

Thank You

DecodeLabs AI-ML Virtual Internship | Artificial Intelligence (AI)

Vansh

Roll No. 25SCS1003000858 | IILM University, Greater Noida